

PART 0: Requirements

Make sure Docker and Docker Compose are installed.

```
docker --version
docker compose version
```

PART 1: Create Project Folder

```
mkdir multi-container-app
cd multi-container-app
mkdir backend frontend
```

PART 2: Backend Setup

Navigate to backend folder and create files.

```
cd backend
touch server.js package.json Dockerfile
```

server.js

```
const express = require("express");
const mongoose = require("mongoose");

const app = express();
app.use(express.json());

mongoose.connect("mongodb://mongo:27017/testdb");

app.get("/", (req, res) => {
  res.send("Backend running 🟢");
});

app.listen(5000);
```

package.json

```
{
  "dependencies": {
    "express": "^4.18.2",
    "mongoose": "^7.0.0"
  }
}
```

Dockerfile

```
FROM node:18
WORKDIR /app
COPY package.json .
RUN npm install
COPY . .
EXPOSE 5000
CMD ["node", "server.js"]
```

PART 3: Frontend Setup

```
cd ../frontend
```

```
touch index.html Dockerfile
```

index.html

```
<h1>Hello from Frontend</h1>
```

Dockerfile

```
FROM nginx:alpine
COPY . /usr/share/nginx/html
```

PART 4: Docker Compose

```
version: "3.9"
services:
  mongo:
    image: mongo
    ports:
      - "27017:27017"

  backend:
    build: ./backend
    ports:
      - "5000:5000"

  frontend:
    build: ./frontend
    ports:
      - "3000:80"
```

PART 5: Run

```
docker compose up --build
```

PART 6: Access

Frontend: <http://localhost:3000>

Backend: <http://localhost:5000>